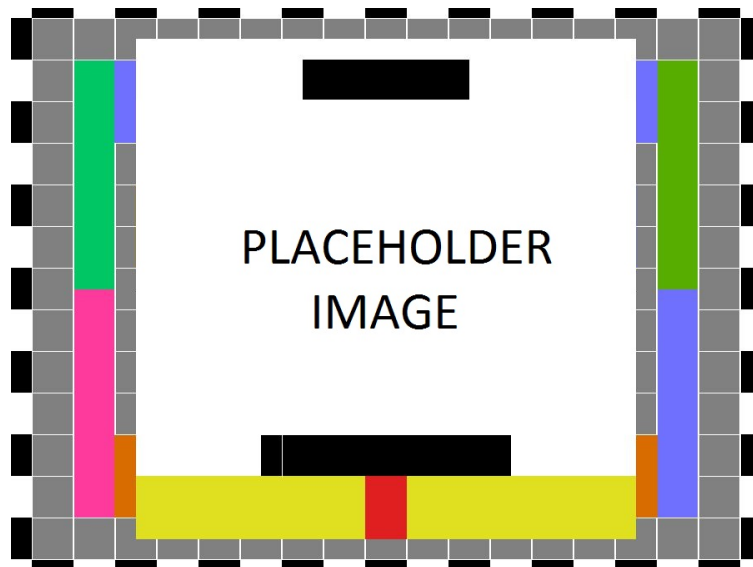


**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
THE UNIVERSITY OF TEXAS AT ARLINGTON**

**PROJECT CHARTER
CSE 4316: SENIOR DESIGN I
FALL 2026**



**TEAM NIMBUS
AQUAGUARD SMART WATER-LEAK MONITOR**

ALAN TURING
GRACE HOPPER
JOHN VON NEUMANN
ADA LOVELACE
CHARLES BABBAGE

REVISION HISTORY

Revision	Date	Author(s)	Description
0.1	09.02.2026	GH	document creation
1.0	09.15.2026	AT, GH, CB	official release

CONTENTS

1	Problem Statement	4
2	Methodology	4
3	Value Proposition	4
4	Development Milestones	4
5	Background	5
6	Related Work	5
7	System Overview	5
8	Project Management Approach	5
9	Roles & Responsibilities	5
10	Cost Proposal	5
11	Facilities & Equipment	5
12	Assumptions	5
13	Constraints	6
14	Risks	6
15	Documentation & Reporting	6

1 PROBLEM STATEMENT

Household water leaks under sinks and behind appliances often go unnoticed for hours or days, causing expensive damage and wasted water. Homeowners have no early warning until they see or smell the damage. What is needed is an inexpensive way to detect leaks the moment they start and alert the homeowner immediately, wherever they are.

2 METHODOLOGY

Team Nimbus will build AquaGuard Smart Water-Leak Monitor, a small battery-powered sensor device that detects moisture and abnormal flow under a sink and communicates over Bluetooth Low Energy (BLE) with a companion phone app. The app pairs and configures the device, raises immediate alerts on a leak, and shows event history. In short: we are building a small sensor plus a phone app that warns you the instant a leak starts.

3 VALUE PROPOSITION

For homeowners, AquaGuard turns costly, late-discovered water damage into an early, actionable alert, at a fraction of the cost of professional leak systems. For the university, the project exercises the full IoT stack — embedded firmware, BLE protocol design, a cross-platform mobile app, and cloud push notifications.

4 DEVELOPMENT MILESTONES

- Project Charter first draft - September 2026
- System Requirements Specification - October 2026
- Architectural Design Specification - November 2026
- Demonstration of BLE pairing & live sensor read - December 2026
- Detailed Design Specification - February 2027
- Demonstration of on-device leak detection & local alert - March 2027
- Demonstration of remote push notification - April 2027
- CoE Innovation Day poster presentation - April 2027
- Final Project Demonstration (end-to-end leak alert) - May 2027

5 BACKGROUND

The project is team-initiated with an industry mentor from a home-automation startup acting as advisor. There is no paying customer; the target user is a general homeowner. Requirements were gathered from mentor interviews and a survey of commercial leak sensors.

6 RELATED WORK

Commercial leak sensors (e.g. Wi-Fi water sensors) exist [1] but many require a hub, mains power, or a subscription. AquaGuard targets a low-cost, battery-powered, hub-free BLE device with an open companion app. (Add ≥ 5 IEEE references in the final version.)

7 SYSTEM OVERVIEW

The sensor device (moisture pad + flow sensor + microcontroller with BLE) advertises readings and events. The phone app (BLE central) pairs with the device, configures thresholds, alerts locally on a leak, and forwards events to a lightweight cloud backend that delivers push notifications when the phone is away. See the ADS for the three-layer design.

8 PROJECT MANAGEMENT APPROACH

Iterative Scrum [1, 2] per ISO/IEC/IEEE 16326:2019 [3] within ISO/IEC/IEEE 12207:2017 [4] and 15288:2023 [5] (the latter covering the hardware element). Two-week sprints; estimation in task-hours; progress by burndown and requirement/test coverage. Git with tagged releases; firmware and app versioned together.

9 ROLES & RESPONSIBILITIES

Advisor: home-automation mentor. Product Owner: Grace Hopper. Scrum Master: Alan Turing. Firmware/hardware: Charles Babbage; BLE protocol: John Von Neumann; Mobile app: Ada Lovelace; Cloud/push: Grace Hopper; Integration/test: Alan Turing.

10 COST PROPOSAL

Major expenses: microcontroller + BLE module, moisture/flow sensors, battery and enclosure (3D-printed), and a small cloud allocation for push. Funded by the default CSE stipend. Largest line item is the sensor + BLE module across several prototype revisions.

11 FACILITIES & EQUIPMENT

Senior Design lab (soldering, 3D printing, bench supply), personal iOS/Android test phones, and a controlled water test rig (a tray and dripper) by the 3rd sprint.

12 ASSUMPTIONS

- Team members can test on both a recent iOS and a recent Android phone.
- A controlled water test rig will be available by the 3rd sprint.
- A free-tier push-notification service will meet demo volume.
- BLE range under a cabinet (≤ 5 m through one wall) is sufficient.
- The device runs ≥ 6 months on one coin/AA battery set.

13 CONSTRAINTS

- Final demonstration must be completed by May 1st, 2027.
- Total development cost must not exceed \$800.
- The device must be battery-powered (no mains wiring).
- The app must run on current iOS and Android without app-store approval delays blocking the demo (use test/ad-hoc builds).
- Wireless design must comply with BLE (Bluetooth SIG) specifications.

14 RISKS

Risk description	Probability	Loss (days)	Exposure (days)
BLE reliability/background behavior differs iOS vs Android	0.50	12	6.0
Battery life target hard to meet	0.40	12	4.8
Sensor false positives/negatives on real leaks	0.40	10	4.0
Push notification delivery when app backgrounded	0.30	8	2.4
Enclosure waterproofing revisions	0.25	8	2.0

Table 2: Highest-exposure project risks

15 DOCUMENTATION & REPORTING

Documentation deliverables in Git, released at their planning sprints; recurring sprint artifacts and engineering notebooks each sprint. Closeout: sensor prototype + app, poster, public web page, demo video, documented source (MIT), firmware sources, enclosure CAD (STL/STEP), wiring schematic, app build/deploy instructions, and a user setup guide.

REFERENCES

- [1] K. S. Rubin, *Essential Scrum: A Practical Guide to the Most Popular Agile Process*, 1st ed. Addison-Wesley Professional, 2012.
- [2] K. Schwaber and J. Sutherland, “The Scrum Guide: The Definitive Guide to Scrum: The Rules of the Game,” 2020.
- [3] ISO/IEC/IEEE, “ISO/IEC/IEEE 16326:2019 — Systems and software engineering — Life cycle processes — Project management,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2019.
- [4] —, “ISO/IEC/IEEE 12207:2017 — Systems and software engineering — Software life cycle processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2017.
- [5] —, “ISO/IEC/IEEE 15288:2023 — Systems and software engineering — System life cycle processes,” International Organization for Standardization, Geneva, Switzerland, Tech. Rep., 2023.